$$Prove\ \{n \in \mathbb{N} \land\ n \geq 3 \Rightarrow n^2 \geq 2n + 3\}$$

$$Base\ Case:\ n = 3$$

$$9 \geq 6 + 3$$

$$9 \geq 9$$

$Proof:\ n + 1$

$n^2 + 2n + 1 \geq 2n + 5$

$Because\ n^2 \geq 2n\ for$
$any\ natural\ integer$
$greater\ than\ 2,$

$And\ 2n \geq 5\ for\ any$
$natural\ integer\ greater$
$than\ 2,$

$the\ sum\ of\ n^2 + 2n$
$will\ be\ greater\ than$
$the\ sum\ of\ 2n + 5$

$Thus\ S(n+1)\ is\ also\ true.$

$Since\ S(n) \Rightarrow S(n+1)\ is\ true,\ all\ positive$
$integers\ n\ that\ are\ greater\ than\ 3\ prove$
$S(n) \Rightarrow S(n+1).$